

KARTIK GUHA

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OBJECTIVE

Accomplished Engineer with good hold on Object Oriented Design, SDLC, Cloud and AI, looking for prospect to work in an organization where my talent, knowledge and skills can be best utilized for the development of the organization; giving me opportunity to increase my knowledge at the same time.

SKILLS

Programming	: Python, GoLang, C#
GUI	: WPF, Windows Forms, QT5
Database	: Oracle, MySQL/MariaDB, MangoDB, Duck DB, Databricks
OS	: Windows, Linux
Frameworks	: Gen AI, Langchain, vLLM, MVC, MVVM Blockchain, SoA
Testing	: Unit Testing, Functional Testing
Other	: Preparing proposals and RFQ responses, Solution Creation, Algorithm development

EXPERIENCE

TATA CONSULTANCY SERVICES LTD. [MARCH 2012 - TILL DATE]

- *[April 2024-Ongoing] Cloud & AI Transformation:* Transforming the companywide work style by introducing AI and unifying Data across various departments.
  - My Role: Solution Architect /Technical Manager
    - Market Research and Ideation
    - Frequent discussion with client to define the strategy
    - Migration of Classic AUTOSAR based ECUs to software component/virtual ECUs
    - Devising strategy for Cloud based over the air updates to vehicle ECUs
    - Project management
- *[April 2023-Ongoing] Gen Ai based Integrated Testcase Creation Platform: Creating a common platform to reuse and dynamically train* the language models for testcase and test script creation for various ECU programs
  - My Role: Solution Architect/Technical Manager
    - Market Research, Architecture Ideation
    - Drafting Strategy for Dataset creation & Dataset normalization
    - Setting up in-house server hosting for hosting LLMs
    - Devising LORA based strategy to change LLM behavior dynamically based on department/ECU program
    - Project management
- *[April 2021- March 2023] EE Architecture*: Migrating vehicle platform to futuristic cloud based flexible architecture that can leverage the data for various analysis tasks and ML training
  - My Role: Technical Manager
    - Frequent discussion with client to define the strategy
    - Migration of traditional Gateway/Domain based architecture to Centralized/Domain computer-based architecture
    - Converting event triggered actions to Service oriented Architecture (SoA)
    - Creating cloud first concept using Databricks

- *[April 2019-March 2021] Intelligent* Life-on-Board Features: Creating PoCs for various upcoming luxury/ convenience features in vehicles including Ml based user aware cockpit (voice/gaze awareness using OpenCV and SpeechRecognition, integrated with smartphone).
  - My Role: Solution Architect/Lead
    - Client handling and business development
    - Requirement gathering and analysis
    - Software design
    - Entity relationship model creation
    - UML model creation
- *[September 2017-March 2019] Crash Data Analyzer*: Creation of software for translating and analyzing the data retrieved from vehicle Event Data Recorder.
  - My Role: Solution Architect/Lead
    - Client handling and business development
    - Requirement gathering and analysis
    - Software design
    - Entity relationship model creation
    - UML model creation
- *[April 2016-August 2017] iKey Spatial Coverage Analyzer*: Finding the blind spots/coverage of key fob inside and outside the vehicle by creating spherical coverage field using extrapolated point cloud data received from the signal detector at different places.
  - My Role: Technical Architect/Lead
    - Client handling
    - Requirement gathering and analysis
    - Entity relationship model creation
    - UML model creation

- *[January 2014-March 2016] Vehicle Data Analysis*: Study/analysis of various aspects related to sales/usage of cars to facilitate the decision making. Key inputs were GPS data, battery SoC data, diagnostic data and sales reports.
  - My Role: Software Architect/Lead
    - Requirement gathering and analysis
    - Entity relationship model creation
    - UML model creation
    - Software development
- *[June 2012-December 2013] Harness Simulation*: Automation of the vehicle harness design testing and creation the V/I analysis reports. This tool also included property editor for components and Automatic resolution of Design errors if found.
  - My Role: Requirement gathering and analysis
    - UML model creation
    - Software development

- *Training Projects [TCS]*:

  1) Postpaid Billing System: A C++ based console application for Unix OS that emulated the billing and employee management of a telecom company. I was awarded for coming up with the best solution for this problem statement.

  2) Oracle Interface for Unix OS: As for above project, we required Oracle as database which and had no interface for Unix; I created a separate application that provided a basic interface for Oracle in Unix OS. I was awarded for my out of the box thinking. The application is still used by the trainees

- *B.E. Projects*

  1) Free SMS Portal: Teamed up with Ericsson Labs to create a SMS portal. This portal enabled the registered users to send free SMS across the globe.

  2) Smart CPU Sharing: A P2P based, research-oriented project to facilitate sharing of loads between computers, connected over LAN. The aim was to monitor the CPU load of all the peers and if the load exceeds the predefined threshold, the CPU will send HELP message to the peers (connected via LAN). If free, the systems will accept the request. The help seeker will then divide the load between those who accepted its request. The pending instructions will be sent to the peers, who then will send the results back.

CARRIER HIGHLIGHTS

- Successfully handled TCS' participation in Automotive World 2023. I was responsible for planning, operations and co-ordination between teams as well as customer connect.
- Created connects with over 10 potential accounts for AI related projects.
- Lean Six Sigma Green Belt Certified for "Using Software for Process Improvement".
- Received client's internal "Shinkansen Award" for drastically reducing the time required for analyzing vehicle crash data.

EDUCATION

- Bachelor of Engineering (Information Technology): Shri Vaishnav Institute of Technology and Science, Indore (SVITS) with 68.24% aggregate.
- Higher Secondary: Harsukh Digamber Higher Secondary School with 80% aggregate.

PERSONAL DETAILS

- Date of Birth : 29th March 1988
- Hobbies : Badminton, reading, playing piano, composing music